Kafka

“A man who doesn't spend time with his family can never be a real man.”
- Mario Puzo, The Godfather

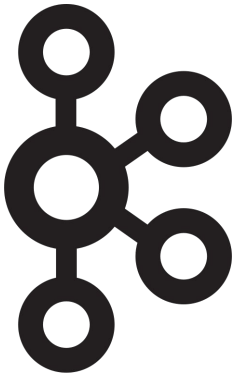


Outline

1. Definitions
2. Use Cases
3. Worst Cases

1.1. Introduction

- Apache Kafka is a **distributed event streaming platform**.
- Kafka can act as a message broker, its capabilities extend beyond traditional message broker such as **high-throughput data streaming** and **long-term data retention**
- Kafka was originally developed at LinkedIn, and was open sourced in 2011.



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A Distributed Event Streaming Platform

1.1. Differences between Response vs Event?

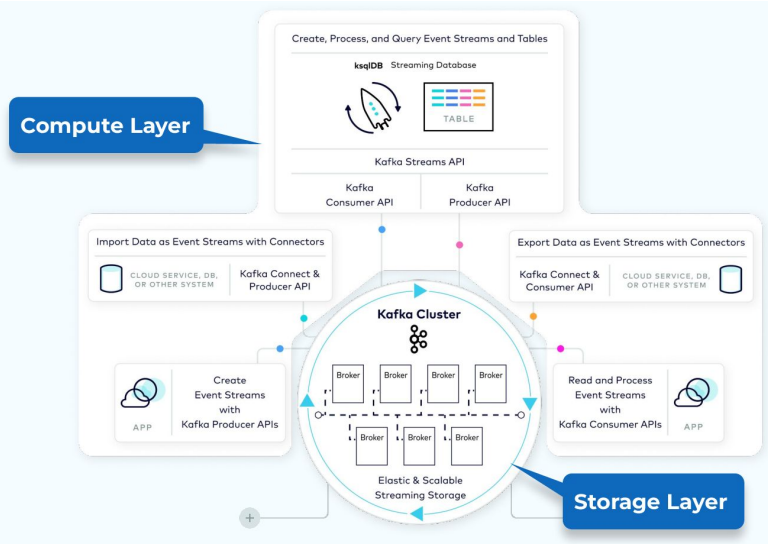
- Response:
  - Be sent to a specific client (that made a request)
  - Variable (over time)
  - Synchronous communication
- **Event**:
  - Just a thing that has happened → **immutable**
  - Business fact that have **value to more than one service**.
  - Asynchronous communication

1.2. Event Streaming

- Streaming: a **continuous flow** of data (event) **in real-time**
- Event streaming:
  - How to get events in/out
  - How to store events
  - Ordering of events
  - …
- Note: Streaming != Stream Processing

1.3. Platform

- Kafka platform provides additional functionalities:
  - Data Integration
  - Stream Processing
  - Ecosystem and Tools, …
- Kafka APIs:
  - **Producer API**
  - **Consumer API**
  - Streams API
  - Connect API
  - AdminClient API



Source: Confluent